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1. A complex number z has modulus 1 and argument θ.

(a) Show that

$$z^n + \frac{1}{z^n} = 2\cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2\mathrm{i}\sin n\theta, \qquad n \in \mathbb{Z}^+$$ **[3]**

(b) Hence, by expanding $\left(z + \frac{1}{z}\right)^3$ and $\left(z - \frac{1}{z}\right)^3$, show that

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$ **[4]**

Solution

(a) Since $|z| = 1$ and $\arg z = \theta$,

$$z = \cos\theta + \mathrm{i}\sin\theta = e^{\mathrm{i}\theta}$$

So, by De Moivre's theorem,

$$z^n = e^{\mathrm{i}n\theta} = \cos n\theta + \mathrm{i}\sin n\theta$$

Also,

$$\frac{1}{z} = e^{-\mathrm{i}\theta}$$

and hence

$$\frac{1}{z^n} = e^{-\mathrm{i}n\theta} = \cos n\theta - \mathrm{i}\sin n\theta$$

Now add:

$$z^n + \frac{1}{z^n} = (\cos n\theta + \mathrm{i}\sin n\theta) + (\cos n\theta - \mathrm{i}\sin n\theta) = 2\cos n\theta$$

Now subtract:

$$z^n - \frac{1}{z^n} = (\cos n\theta + \mathrm{i}\sin n\theta) - (\cos n\theta - \mathrm{i}\sin n\theta) = 2\mathrm{i}\sin n\theta$$

Therefore

$$z^n + \frac{1}{z^n} = 2\cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2\mathrm{i}\sin n\theta$$

(b) From part (a), with $n = 1$,

$$z + \frac{1}{z} = 2\cos\theta$$

Cubing both sides gives

$$\left(z + \frac{1}{z}\right)^3 = (2\cos\theta)^3 = 8\cos^3\theta$$

Expand the left-hand side:

$$\begin{aligned}\left(z + \frac{1}{z}\right)^3 &= z^3 + 3z + \frac{3}{z} + \frac{1}{z^3} \\ &= \left(z^3 + \frac{1}{z^3}\right) + 3\left(z + \frac{1}{z}\right)\end{aligned}$$

Using part (a) again,

$$8\cos^3\theta = 2\cos 3\theta + 6\cos\theta$$

So

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

Now use part (a) with

$$z - \frac{1}{z} = 2\mathrm{i}\sin\theta$$

Cubing both sides,

$$\left(z - \frac{1}{z}\right)^3 = (2\mathrm{i}\sin\theta)^3 = -8\mathrm{i}\sin^3\theta$$

Expand:

$$\begin{aligned}\left(z - \frac{1}{z}\right)^3 &= z^3 - 3z + \frac{3}{z} - \frac{1}{z^3} \\ &= \left(z^3 - \frac{1}{z^3}\right) - 3\left(z - \frac{1}{z}\right)\end{aligned}$$

Using part (a),

$$-8\mathrm{i}\sin^3\theta = 2\mathrm{i}\sin 3\theta - 6\mathrm{i}\sin\theta$$

Hence

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

Let $t = \tan\theta$. Then

$$\begin{aligned}\tan 3\theta &= \frac{\sin 3\theta}{\cos 3\theta} \\ &= \frac{\dfrac{\sin 3\theta}{\cos^3\theta}}{\dfrac{\cos 3\theta}{\cos^3\theta}} \\ &= \frac{\dfrac{3\sin\theta - 4\sin^3\theta}{\cos^3\theta}}{\dfrac{4\cos^3\theta - 3\cos\theta}{\cos^3\theta}} \\ &= \frac{3\tan\theta\sec^2\theta - 4\tan^3\theta}{4 - 3\sec^2\theta} \\ &= \frac{3t(1+t^2) - 4t^3}{4 - 3(1+t^2)} \\ &= \frac{3t - t^3}{1 - 3t^2}\end{aligned}$$

Therefore

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

2. **(a)** Using de Moivre's theorem, show that

$$\sin^4\theta\cos^2\theta \equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$$ **[5]**

(b) By using the substitution $\alpha = \left(\frac{\pi}{2} - \theta\right)$, or otherwise, find a similar identity for $\sin^2\theta\cos^4\theta$. **[3]**

(c) Given that $0 < a < \frac{\pi}{2}$ and

$$\int_0^a \left(\sin^4\theta\cos^2\theta + 2\sin^2\theta\cos^4\theta\right)\,\mathrm{d}\theta = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$$

find the exact value of a. **[5]**

Solution

(a) Let

$$c = \cos 2\theta$$

Using the double-angle identities,

$$\sin^2\theta = \frac{1-\cos 2\theta}{2} = \frac{1-c}{2}, \qquad \cos^2\theta = \frac{1+\cos 2\theta}{2} = \frac{1+c}{2}$$

So

$$\begin{aligned}\sin^4\theta\cos^2\theta &= \left(\frac{1-c}{2}\right)^2\left(\frac{1+c}{2}\right)\\ &= \frac{1}{8}(1-2c+c^2)(1+c)\\ &= \frac{1}{8}(1+c-2c-2c^2+c^2+c^3)\\ &= \frac{1}{8}(1-c-c^2+c^3)\end{aligned}$$

Now use de Moivre's theorem.
From

$$(\cos 2\theta + \mathrm{i}\sin 2\theta)^2 = \cos 4\theta + \mathrm{i}\sin 4\theta$$

taking real parts gives

$$\cos 4\theta = 2\cos^2 2\theta - 1 = 2c^2 - 1$$

so

$$c^2 = \frac{1+\cos 4\theta}{2}$$

Also, from

$$(\cos 2\theta + \mathrm{i}\sin 2\theta)^3 = \cos 6\theta + \mathrm{i}\sin 6\theta$$

taking real parts gives

$$\cos 6\theta = 4\cos^3 2\theta - 3\cos 2\theta = 4c^3 - 3c$$

so

$$c^3 = \frac{\cos 6\theta + 3c}{4}$$

Substituting these into $\frac{1}{8}(1-c-c^2+c^3)$,

$$\begin{aligned}\sin^4\theta\cos^2\theta &= \frac{1}{8}\left(1 - c - \frac{1+\cos 4\theta}{2} + \frac{\cos 6\theta + 3c}{4}\right)\\ &= \frac{1}{8}\left(\frac{4-4c-2-2\cos 4\theta+\cos 6\theta+3c}{4}\right)\\ &= \frac{1}{32}\left(2-c-2\cos 4\theta+\cos 6\theta\right)\end{aligned}$$

Since $c = \cos 2\theta$,

$$\sin^4\theta\cos^2\theta \equiv \frac{1}{32}\left(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2\right)$$

Hence the required identity is shown.

(b) Let

$$\alpha = \frac{\pi}{2} - \theta$$

Then

$$\sin\theta = \cos\alpha, \qquad \cos\theta = \sin\alpha$$

so

$$\sin^2\theta\cos^4\theta = \cos^2\alpha\sin^4\alpha = \sin^4\alpha\cos^2\alpha$$

Using part (a),

$$\sin^4\alpha\cos^2\alpha = \frac{1}{32}\left(\cos 6\alpha - 2\cos 4\alpha - \cos 2\alpha + 2\right)$$

Now

$$2\alpha = \pi - 2\theta, \quad 4\alpha = 2\pi - 4\theta, \quad 6\alpha = 3\pi - 6\theta$$

so

$$\cos 2\alpha = -\cos 2\theta, \qquad \cos 4\alpha = \cos 4\theta, \qquad \cos 6\alpha = -\cos 6\theta$$

Therefore

$$\sin^2\theta\cos^4\theta = \frac{1}{32}\left(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2\right)$$

So a similar identity is

$$\sin^2\theta\cos^4\theta \equiv \frac{1}{32}\left(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2\right)$$

(c) Let

$$I(a) = \int_0^a \left(\sin^4\theta\cos^2\theta + 2\sin^2\theta\cos^4\theta\right)\,\mathrm{d}\theta$$

Using parts (a) and (b),

$$\begin{aligned}\sin^4\theta\cos^2\theta + 2\sin^2\theta\cos^4\theta &= \frac{1}{32}\left(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2\right)\\ &\quad + \frac{2}{32}\left(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2\right)\\ &= \frac{1}{32}\left(-\cos 6\theta - 6\cos 4\theta + \cos 2\theta + 6\right)\end{aligned}$$

Hence

$$\begin{aligned}I(a) &= \frac{1}{32}\int_0^a \left(-\cos 6\theta - 6\cos 4\theta + \cos 2\theta + 6\right)\,\mathrm{d}\theta\\ &= \frac{1}{32}\left[-\frac{1}{6}\sin 6\theta - \frac{3}{2}\sin 4\theta + \frac{1}{2}\sin 2\theta + 6\theta\right]_0^a\\ &= \frac{1}{32}\left(6a + \frac{1}{2}\sin 2a - \frac{3}{2}\sin 4a - \frac{1}{6}\sin 6a\right)\end{aligned}$$

We are given that

$$I(a) = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$$

Try $a = \frac{\pi}{6}$. Then

$$6a = \pi, \qquad \sin 2a = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}, \qquad \sin 4a = \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}, \qquad \sin 6a = \sin\pi = 0$$

So

$$\begin{aligned} I\left(\frac{\pi}{6}\right) &= \frac{1}{32}\left(\pi + \frac{1}{2}\cdot\frac{\sqrt{3}}{2} - \frac{3}{2}\cdot\frac{\sqrt{3}}{2} - 0\right) \\ &= \frac{1}{32}\left(\pi - \frac{\sqrt{3}}{2}\right) \\ &= \frac{\pi}{32} - \frac{\sqrt{3}}{64} \end{aligned}$$

So $a = \frac{\pi}{6}$ satisfies the equation.
To show this is the only solution, differentiate $I(a)$ using the Fundamental Theorem of Calculus:

$$I'(a) = \sin^4 a \cos^2 a + 2\sin^2 a \cos^4 a$$

and factorise:

$$I'(a) = \sin^2 a \cos^2 a(\sin^2 a + 2\cos^2 a)$$

For $0 < a < \frac{\pi}{2}$, we have $\sin a > 0$ and $\cos a > 0$, so

$$I'(a) > 0$$

Thus $I(a)$ is strictly increasing on $\left(0, \frac{\pi}{2}\right)$, so the solution is unique.
Therefore

$$a = \frac{\pi}{6}$$

that is, $a = 30°$.

3. Use de Moivre's theorem to find the constants A, B and C in the identity

$$\sin\theta\cos^4\theta = A\sin\theta + B\sin 3\theta + C\sin 5\theta$$ **[4]**

Solution

Let

$$z = \cos\theta + \mathrm{i}\sin\theta$$

Then $z^{-1} = \cos\theta - \mathrm{i}\sin\theta$, so

$$\cos\theta = \frac{z+z^{-1}}{2}, \qquad \sin\theta = \frac{z-z^{-1}}{2\mathrm{i}}$$

Also, by de Moivre's theorem,

$$z^n = \cos n\theta + \mathrm{i}\sin n\theta$$

which gives

$$\sin n\theta = \frac{z^n - z^{-n}}{2\mathrm{i}}$$

Now write $\sin\theta\cos^4\theta$ in terms of z:

$$\begin{aligned}\sin\theta\cos^4\theta &= \frac{z-z^{-1}}{2\mathrm{i}}\left(\frac{z+z^{-1}}{2}\right)^4\\ &= \frac{1}{32\mathrm{i}}(z-z^{-1})(z+z^{-1})^4\end{aligned}$$

Expand $(z+z^{-1})^4$:

$$(z+z^{-1})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$$

So

$$\begin{aligned}\sin\theta\cos^4\theta &= \frac{1}{32\mathrm{i}}(z-z^{-1})(z^4+4z^2+6+4z^{-2}+z^{-4})\\ &= \frac{1}{32\mathrm{i}}\big(z^5+4z^3+6z+4z^{-1}+z^{-3}\\ &\qquad\qquad - z^3 - 4z - 6z^{-1} - 4z^{-3} - z^{-5}\big)\\ &= \frac{1}{32\mathrm{i}}\big(z^5+3z^3+2z-2z^{-1}-3z^{-3}-z^{-5}\big)\\ &= \frac{1}{32\mathrm{i}}\Big[(z^5-z^{-5})+3(z^3-z^{-3})+2(z-z^{-1})\Big]\end{aligned}$$

Now use

$$\frac{z^n - z^{-n}}{2\mathrm{i}} = \sin n\theta$$

Hence

$$\begin{aligned}\sin\theta\cos^4\theta &= \frac{1}{16}\sin 5\theta + \frac{3}{16}\sin 3\theta + \frac{2}{16}\sin\theta\\ &= \frac{1}{8}\sin\theta + \frac{3}{16}\sin 3\theta + \frac{1}{16}\sin 5\theta\end{aligned}$$

Comparing with

$$\sin\theta\cos^4\theta = A\sin\theta + B\sin 3\theta + C\sin 5\theta$$

gives

$$A = \frac{1}{8}, \qquad B = \frac{3}{16}, \qquad C = \frac{1}{16}$$

4. **(i)** Starting with the result

$$e^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$$

show that

(A)

$$(\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta \qquad \textbf{[2]}$$

(B)

$$\sin\theta = \frac{1}{2\mathrm{i}}\left(e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}\right) \qquad \textbf{[2]}$$

(ii) Using the result in part (i)(A), obtain the values of the constants a, b, c, d and e in the identity

$$\cos 8\theta \equiv a\sin^8\theta + b\sin^6\theta + c\sin^4\theta + d\sin^2\theta + e \qquad \textbf{[6]}$$

(iii) Using the result in part (i)(B), obtain the values of the constants P, Q, R, S and T in the identity

$$\sin^8\theta \equiv P\cos 8\theta + Q\cos 6\theta + R\cos 4\theta + S\cos 2\theta + T \qquad \textbf{[6]}$$

(iv) Hence show that

$$\sin\frac{\pi}{8} = \left(\frac{17 - 12\sqrt{2}}{64}\right)^{\frac{1}{8}} \qquad \textbf{[3]}$$

Solution

(i) **(A)** Starting from

$$e^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$$

raise both sides to the power n:

$$(\cos\theta + \mathrm{i}\sin\theta)^n = \left(e^{\mathrm{i}\theta}\right)^n = e^{\mathrm{i}n\theta}$$

Now apply Euler's formula again, with θ replaced by $n\theta$:

$$e^{\mathrm{i}n\theta} = \cos n\theta + \mathrm{i}\sin n\theta$$

Hence

$$(\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$$

(B) Replace θ by $-\theta$ in Euler's formula:

$$e^{-\mathrm{i}\theta} = \cos(-\theta) + \mathrm{i}\sin(-\theta) = \cos\theta - \mathrm{i}\sin\theta$$

Subtract this from

$$e^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$$

to get

$$e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta} = 2\mathrm{i}\sin\theta$$

Therefore

$$\sin\theta = \frac{1}{2\mathrm{i}}\left(e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}\right)$$

(ii) Let $C = \cos\theta$ and $S = \sin\theta$. From part **(i)(A)**,

$$(C + \mathrm{i}S)^8 = \cos 8\theta + \mathrm{i}\sin 8\theta$$

Expand $(C + \mathrm{i}S)^8$:

$$\begin{aligned}(C + \mathrm{i}S)^8 &= C^8 + 8\mathrm{i}C^7S + 28C^6(\mathrm{i}S)^2 + 56C^5(\mathrm{i}S)^3 + 70C^4(\mathrm{i}S)^4 \\ &\quad + 56C^3(\mathrm{i}S)^5 + 28C^2(\mathrm{i}S)^6 + 8C(\mathrm{i}S)^7 + (\mathrm{i}S)^8 \\ &= C^8 + 8\mathrm{i}C^7S - 28C^6S^2 - 56\mathrm{i}C^5S^3 + 70C^4S^4 \\ &\quad + 56\mathrm{i}C^3S^5 - 28C^2S^6 - 8\mathrm{i}CS^7 + S^8\end{aligned}$$

Taking real parts,

$$\cos 8\theta = C^8 - 28C^6S^2 + 70C^4S^4 - 28C^2S^6 + S^8$$

Since $C^2 = 1 - S^2$, let $x = S^2$. Then

$$\begin{aligned}\cos 8\theta &= (1-x)^4 - 28(1-x)^3x + 70(1-x)^2x^2 - 28(1-x)x^3 + x^4 \\ &= (1 - 4x + 6x^2 - 4x^3 + x^4) \\ &\quad - 28(x - 3x^2 + 3x^3 - x^4) \\ &\quad + 70(x^2 - 2x^3 + x^4) - 28(x^3 - x^4) + x^4 \\ &= 1 - 32x + 160x^2 - 256x^3 + 128x^4\end{aligned}$$

Substitute back $x = \sin^2\theta$:

$$\cos 8\theta \equiv 128\sin^8\theta - 256\sin^6\theta + 160\sin^4\theta - 32\sin^2\theta + 1$$

Hence

$$a = 128, \quad b = -256, \quad c = 160, \quad d = -32, \quad e = 1$$

(iii) From part **(i)(B)**,

$$\sin\theta = \frac{1}{2\mathrm{i}}\left(e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}\right)$$

Therefore

$$\sin^8\theta = \left(\frac{1}{2\mathrm{i}}\right)^8\left(e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}\right)^8 = \frac{1}{256}\left(e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}\right)^8$$

Now expand:

$$\begin{aligned}\sin^8\theta = \frac{1}{256}\Big(&e^{8\mathrm{i}\theta} - 8e^{6\mathrm{i}\theta} + 28e^{4\mathrm{i}\theta} - 56e^{2\mathrm{i}\theta} + 70 \\ &- 56e^{-2\mathrm{i}\theta} + 28e^{-4\mathrm{i}\theta} - 8e^{-6\mathrm{i}\theta} + e^{-8\mathrm{i}\theta}\Big)\end{aligned}$$

Use

$$e^{k\mathrm{i}\theta} + e^{-k\mathrm{i}\theta} = 2\cos k\theta$$

so that

$$\begin{aligned}\sin^8\theta &= \frac{1}{256}\Big((e^{8\mathrm{i}\theta} + e^{-8\mathrm{i}\theta}) - 8(e^{6\mathrm{i}\theta} + e^{-6\mathrm{i}\theta}) \\ &\qquad + 28(e^{4\mathrm{i}\theta} + e^{-4\mathrm{i}\theta}) - 56(e^{2\mathrm{i}\theta} + e^{-2\mathrm{i}\theta}) + 70\Big) \\ &= \frac{1}{256}\Big(2\cos 8\theta - 16\cos 6\theta + 56\cos 4\theta - 112\cos 2\theta + 70\Big) \\ &= \frac{1}{128}\cos 8\theta - \frac{1}{16}\cos 6\theta + \frac{7}{32}\cos 4\theta - \frac{7}{16}\cos 2\theta + \frac{35}{128}\end{aligned}$$

So

$$P = \frac{1}{128}, \quad Q = -\frac{1}{16}, \quad R = \frac{7}{32}, \quad S = -\frac{7}{16}, \quad T = \frac{35}{128}$$

Equivalently,

$$\sin^8\theta \equiv \frac{\cos 8\theta - 8\cos 6\theta + 28\cos 4\theta - 56\cos 2\theta + 35}{128}$$

(iv) Using the identity from part **(iii)** with $\theta = \dfrac{\pi}{8}$,

$$\sin^8\frac{\pi}{8} = \frac{\cos\pi - 8\cos\frac{3\pi}{4} + 28\cos\frac{\pi}{2} - 56\cos\frac{\pi}{4} + 35}{128}$$

Now

$$\cos\pi = -1, \qquad \cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \qquad \cos\frac{\pi}{2} = 0, \qquad \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

so

$$\begin{aligned}\sin^8\frac{\pi}{8} &= \frac{-1-8\left(-\frac{\sqrt{2}}{2}\right)+28(0)-56\left(\frac{\sqrt{2}}{2}\right)+35}{128}\\ &= \frac{-1+4\sqrt{2}-28\sqrt{2}+35}{128}\\ &= \frac{34-24\sqrt{2}}{128}\\ &= \frac{17-12\sqrt{2}}{64}\end{aligned}$$

Since $0<\frac{\pi}{8}<\frac{\pi}{2}$, $\sin\frac{\pi}{8}>0$. Hence

$$\sin\frac{\pi}{8}=\left(\frac{17-12\sqrt{2}}{64}\right)^{\frac{1}{8}}$$

5. **(a)** Use de Moivre's theorem to show that $\cos 6\theta \equiv (1 - 2\sin^2\theta)(16\sin^4\theta - 16\sin^2\theta + 1)$ **[5]**

(b) By solving the equation $\cos 6\theta = 0$, deduce that $\sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$. **[4]**

(c) Hence, or otherwise, write down the exact values of $\sin^2\left(\frac{5\pi}{12}\right)$, $\sin^2\left(\frac{7\pi}{12}\right)$ and $\sin^2\left(\frac{11\pi}{12}\right)$. **[3]**

Solution

(a) Using de Moivre's theorem,

$$(\cos\theta + \mathrm{i}\sin\theta)^6 = \cos 6\theta + \mathrm{i}\sin 6\theta$$

Expand the left-hand side by the binomial theorem:

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)^6 &= \cos^6\theta + 6\cos^5\theta(\mathrm{i}\sin\theta) + 15\cos^4\theta(\mathrm{i}\sin\theta)^2 \\ &\quad + 20\cos^3\theta(\mathrm{i}\sin\theta)^3 + 15\cos^2\theta(\mathrm{i}\sin\theta)^4 \\ &\quad + 6\cos\theta(\mathrm{i}\sin\theta)^5 + (\mathrm{i}\sin\theta)^6 \\ &= \cos^6\theta + 6\mathrm{i}\cos^5\theta\sin\theta - 15\cos^4\theta\sin^2\theta \\ &\quad - 20\mathrm{i}\cos^3\theta\sin^3\theta + 15\cos^2\theta\sin^4\theta \\ &\quad + 6\mathrm{i}\cos\theta\sin^5\theta - \sin^6\theta\end{aligned}$$

Taking real parts gives

$$\cos 6\theta = \cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta$$

Now let

$$u = \sin^2\theta \qquad \text{so} \qquad \cos^2\theta = 1 - u$$

Then

$$\begin{aligned}\cos 6\theta &= (1-u)^3 - 15(1-u)^2u + 15(1-u)u^2 - u^3 \\ &= (1 - 3u + 3u^2 - u^3) - 15(u - 2u^2 + u^3) + 15(u^2 - u^3) - u^3 \\ &= 1 - 3u + 3u^2 - u^3 - 15u + 30u^2 - 15u^3 + 15u^2 - 15u^3 - u^3 \\ &= 1 - 18u + 48u^2 - 32u^3\end{aligned}$$

Now factorise:

$$1 - 18u + 48u^2 - 32u^3 = (1-2u)(16u^2 - 16u + 1)$$

Substituting back $u = \sin^2\theta$,

$$\cos 6\theta = (1 - 2\sin^2\theta)(16\sin^4\theta - 16\sin^2\theta + 1)$$

Hence

$$\cos 6\theta \equiv (1 - 2\sin^2\theta)(16\sin^4\theta - 16\sin^2\theta + 1)$$

(b) Let

$$u = \sin^2\theta$$

From part (a), if $\cos 6\theta = 0$, then

$$(1-2u)(16u^2 - 16u + 1) = 0$$

So either

$$1 - 2u = 0 \quad \Rightarrow \quad u = \frac{1}{2}$$

or

$$\begin{aligned} 16u^2 - 16u + 1 &= 0 \\ u &= \frac{16 \pm \sqrt{(-16)^2 - 4(16)(1)}}{2 \cdot 16} \\ &= \frac{16 \pm \sqrt{256 - 64}}{32} \\ &= \frac{16 \pm \sqrt{192}}{32} \\ &= \frac{16 \pm 8\sqrt{3}}{32} \\ &= \frac{2 \pm \sqrt{3}}{4} \end{aligned}$$

Now

$$\cos\left(6 \cdot \frac{\pi}{12}\right) = \cos\frac{\pi}{2} = 0$$

so $\sin^2\left(\frac{\pi}{12}\right)$ must be one of

$$\frac{1}{2}, \quad \frac{2+\sqrt{3}}{4}, \quad \frac{2-\sqrt{3}}{4}$$

But

$$0 < \frac{\pi}{12} < \frac{\pi}{4}$$

so

$$\sin^2\left(\frac{\pi}{12}\right) < \sin^2\left(\frac{\pi}{4}\right) = \frac{1}{2}$$

Therefore the correct value is

$$\sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$$

(c) Using part (b),

$$\begin{aligned} \sin^2\left(\frac{5\pi}{12}\right) &= \cos^2\left(\frac{\pi}{12}\right) \\ &= 1 - \sin^2\left(\frac{\pi}{12}\right) \\ &= 1 - \frac{2-\sqrt{3}}{4} \\ &= \frac{2+\sqrt{3}}{4} \end{aligned}$$

Also,

$$\sin^2\left(\frac{7\pi}{12}\right) = \sin^2\left(\pi - \frac{5\pi}{12}\right) = \sin^2\left(\frac{5\pi}{12}\right) = \frac{2+\sqrt{3}}{4}$$

and

$$\sin^2\left(\frac{11\pi}{12}\right) = \sin^2\left(\pi - \frac{\pi}{12}\right) = \sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$$

Hence

$$\sin^2\left(\frac{5\pi}{12}\right) = \frac{2+\sqrt{3}}{4}, \qquad \sin^2\left(\frac{7\pi}{12}\right) = \frac{2+\sqrt{3}}{4}, \qquad \sin^2\left(\frac{11\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$$

6. Let $t = \tan\theta$. By writing

$$\cos\theta + \mathrm{i}\sin\theta = \cos\theta\,(1 + \mathrm{i}t)$$

and expanding $(1 + \mathrm{i}t)^6$, use de Moivre's theorem to show that

$$\tan 6\theta = \frac{6\tan\theta - 20\tan^3\theta + 6\tan^5\theta}{1 - 15\tan^2\theta + 15\tan^4\theta - \tan^6\theta}$$ **[7]**

Solution

Let

$$z = \cos\theta + \mathrm{i}\sin\theta$$

and let

$$t = \tan\theta = \frac{\sin\theta}{\cos\theta}$$

Then $\sin\theta = t\cos\theta$, so

$$z = \cos\theta + \mathrm{i}\sin\theta = \cos\theta + \mathrm{i}(t\cos\theta) = \cos\theta(1 + \mathrm{i}t)$$

Now apply de Moivre's theorem:

$$z^6 = \cos 6\theta + \mathrm{i}\sin 6\theta$$

But also, from $z = \cos\theta(1 + \mathrm{i}t)$,

$$z^6 = \cos^6\theta\,(1 + \mathrm{i}t)^6$$

So we expand $(1 + \mathrm{i}t)^6$ using the binomial theorem:

$$(1 + \mathrm{i}t)^6 = 1 + 6(\mathrm{i}t) + 15(\mathrm{i}t)^2 + 20(\mathrm{i}t)^3 + 15(\mathrm{i}t)^4 + 6(\mathrm{i}t)^5 + (\mathrm{i}t)^6$$

Now simplify the powers of i:

$$\mathrm{i}^2 = -1, \qquad \mathrm{i}^3 = -\mathrm{i}, \qquad \mathrm{i}^4 = 1, \qquad \mathrm{i}^5 = \mathrm{i}, \qquad \mathrm{i}^6 = -1$$

Hence

$$\begin{aligned}(1 + \mathrm{i}t)^6 &= 1 + 6\mathrm{i}t + 15(-1)t^2 + 20(-\mathrm{i})t^3 + 15(1)t^4 + 6(\mathrm{i})t^5 + (-1)t^6 \\ &= 1 + 6\mathrm{i}t - 15t^2 - 20\mathrm{i}t^3 + 15t^4 + 6\mathrm{i}t^5 - t^6 \\ &= (1 - 15t^2 + 15t^4 - t^6) + \mathrm{i}(6t - 20t^3 + 6t^5)\end{aligned}$$

Therefore

$$\cos 6\theta + \mathrm{i}\sin 6\theta = \cos^6\theta\Big[(1 - 15t^2 + 15t^4 - t^6) + \mathrm{i}(6t - 20t^3 + 6t^5)\Big]$$

Equating real and imaginary parts gives

$$\cos 6\theta = \cos^6\theta(1 - 15t^2 + 15t^4 - t^6)$$

and

$$\sin 6\theta = \cos^6\theta(6t - 20t^3 + 6t^5)$$

So

$$\begin{aligned}\tan 6\theta &= \frac{\sin 6\theta}{\cos 6\theta} \\ &= \frac{\cos^6\theta(6t - 20t^3 + 6t^5)}{\cos^6\theta(1 - 15t^2 + 15t^4 - t^6)} \\ &= \frac{6t - 20t^3 + 6t^5}{1 - 15t^2 + 15t^4 - t^6}\end{aligned}$$

Since $t = \tan\theta$, this becomes

$$\tan 6\theta = \frac{6\tan\theta - 20\tan^3\theta + 6\tan^5\theta}{1 - 15\tan^2\theta + 15\tan^4\theta - \tan^6\theta}$$

which is the required result.

7. **(a)** Use de Moivre's theorem to show that

$$\sin 6x = \sin 2x \left(a\cos^4 x + b\cos^2 x + c\right)$$

where a, b and c are integers to be determined. **[4]**

(b) Hence solve, for $0 < \theta < \frac{\pi}{2}$,

$$\sin 6\theta = \sin 2\theta \cos 2\theta$$

giving your answers to 3 decimal places. **[4]**

Solution

(a) Let $C = \cos x$ and $S = \sin x$.
By de Moivre's theorem,

$$(\cos x + \mathrm{i}\sin x)^6 = \cos 6x + \mathrm{i}\sin 6x$$

So

$$\begin{aligned}(C+\mathrm{i}S)^6 &= C^6 + 6\mathrm{i}C^5S + 15C^4(\mathrm{i}S)^2 + 20C^3(\mathrm{i}S)^3 + 15C^2(\mathrm{i}S)^4 + 6C(\mathrm{i}S)^5 + (\mathrm{i}S)^6 \\ &= C^6 + 6\mathrm{i}C^5S - 15C^4S^2 - 20\mathrm{i}C^3S^3 + 15C^2S^4 + 6\mathrm{i}CS^5 - S^6\end{aligned}$$

Taking imaginary parts gives

$$\begin{aligned}\sin 6x &= 6C^5S - 20C^3S^3 + 6CS^5 \\ &= 2CS\left(3C^4 - 10C^2S^2 + 3S^4\right)\end{aligned}$$

Now simplify the bracket:

$$\begin{aligned}3C^4 - 10C^2S^2 + 3S^4 &= 3(C^4+S^4) - 10C^2S^2 \\ &= 3\big((C^2+S^2)^2 - 2C^2S^2\big) - 10C^2S^2 \\ &= 3(1-2C^2S^2) - 10C^2S^2 \\ &= 3 - 16C^2S^2 \\ &= 3 - 16C^2(1-C^2) \\ &= 16C^4 - 16C^2 + 3\end{aligned}$$

Also, $\sin 2x = 2\sin x\cos x = 2CS$, so

$$\sin 6x = \sin 2x\left(16\cos^4 x - 16\cos^2 x + 3\right)$$

Hence

$$a = 16, \qquad b = -16, \qquad c = 3$$

(b) Using part (a),

$$\sin 6\theta = \sin 2\theta\left(16\cos^4\theta - 16\cos^2\theta + 3\right)$$

So the equation

$$\sin 6\theta = \sin 2\theta\cos 2\theta$$

becomes

$$\sin 2\theta\left(16\cos^4\theta - 16\cos^2\theta + 3\right) = \sin 2\theta\cos 2\theta$$

Since $0 < \theta < \frac{\pi}{2}$, we have $0 < 2\theta < \pi$, so $\sin 2\theta \neq 0$.
Therefore we can divide by $\sin 2\theta$:

$$16\cos^4\theta - 16\cos^2\theta + 3 = \cos 2\theta$$

Let

$$u = \cos^2\theta$$

Then $\cos 2\theta = 2u - 1$, so

$$\begin{aligned} 16u^2 - 16u + 3 &= 2u - 1 \\ 16u^2 - 18u + 4 &= 0 \end{aligned}$$

Solving,

$$\begin{aligned} u &= \frac{18 \pm \sqrt{18^2 - 4(16)(4)}}{2(16)} \\ &= \frac{18 \pm \sqrt{324 - 256}}{32} \\ &= \frac{18 \pm \sqrt{68}}{32} \\ &= \frac{9 \pm \sqrt{17}}{16} \end{aligned}$$

Hence

$$\begin{aligned} \cos 2\theta &= 2u - 1 \\ &= 2\left(\frac{9 \pm \sqrt{17}}{16}\right) - 1 \\ &= \frac{1 \pm \sqrt{17}}{8} \end{aligned}$$

So

$$2\theta = \arccos\left(\frac{1 \pm \sqrt{17}}{8}\right)$$

and therefore

$$\theta = \frac{1}{2}\arccos\left(\frac{1 \pm \sqrt{17}}{8}\right)$$

This gives

$$\theta \approx 0.438,\ 0.986$$

So the solutions are

$$\theta = 0.438 \text{ rad}, \quad 0.986 \text{ rad}$$

to 3 decimal places.

8. **(a)** By considering the binomial expansion of $(z - z^{-1})^6$, where $z = \cos\theta + \mathrm{i}\sin\theta$, use de Moivre's theorem to show that

$$\sin^6\theta = \frac{1}{32}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big)$$ **[5]**

(b) Hence, using the substitution $x = \tan\theta$, find the exact value of

$$\int_0^1 \frac{x^6}{(1+x^2)^4}\,\mathrm{d}x$$ **[4]**

Solution

(a) Since

$$z = \cos\theta + \mathrm{i}\sin\theta = e^{\mathrm{i}\theta}$$

we have

$$z^{-1} = e^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta$$

and so

$$z - z^{-1} = e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta} = 2\mathrm{i}\sin\theta$$

Hence

$$(z - z^{-1})^6 = (2\mathrm{i}\sin\theta)^6 = 64\mathrm{i}^6\sin^6\theta = -64\sin^6\theta$$

Now expand $(z - z^{-1})^6$ by the binomial theorem:

$$\begin{aligned}(z - z^{-1})^6 &= z^6 - 6z^5(z^{-1}) + 15z^4(z^{-1})^2 - 20z^3(z^{-1})^3 \\ &\quad + 15z^2(z^{-1})^4 - 6z(z^{-1})^5 + (z^{-1})^6 \\ &= z^6 - 6z^4 + 15z^2 - 20 + 15z^{-2} - 6z^{-4} + z^{-6}\end{aligned}$$

By de Moivre's theorem,

$$z^n = \cos n\theta + \mathrm{i}\sin n\theta, \qquad z^{-n} = \cos n\theta - \mathrm{i}\sin n\theta$$

so

$$z^n + z^{-n} = 2\cos n\theta$$

Therefore

$$\begin{aligned}(z - z^{-1})^6 &= (z^6 + z^{-6}) - 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) - 20 \\ &= 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20\end{aligned}$$

Equating this with the earlier result gives

$$-64\sin^6\theta = 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$$

so

$$64\sin^6\theta = 20 - 30\cos 2\theta + 12\cos 4\theta - 2\cos 6\theta$$

and hence

$$\sin^6\theta = \frac{1}{32}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big)$$

This is the required result.

(b) Let

$$I = \int_0^1 \frac{x^6}{(1+x^2)^4}\,\mathrm{d}x$$

Use the substitution

$$x = \tan\theta$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \sec^2\theta \qquad \Rightarrow \qquad \mathrm{d}x = \sec^2\theta\,\mathrm{d}\theta$$

and

$$1 + x^2 = 1 + \tan^2\theta = \sec^2\theta$$

Also, when $x = 0$, $\theta = 0$, and when $x = 1$, $\theta = \frac{\pi}{4}$.
So

$$\begin{aligned} I &= \int_0^{\pi/4} \frac{\tan^6\theta}{(\sec^2\theta)^4}\sec^2\theta\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \tan^6\theta\,\sec^{-6}\theta\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \frac{\sin^6\theta}{\cos^6\theta}\cos^6\theta\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \sin^6\theta\,\mathrm{d}\theta \end{aligned}$$

Using the result from part (a),

$$\sin^6\theta = \frac{1}{32}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big)$$

Therefore

$$\begin{aligned} I &= \frac{1}{32}\int_0^{\pi/4} \big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big)\,\mathrm{d}\theta \\ &= \frac{1}{32}\left[10\theta - \frac{15}{2}\sin 2\theta + \frac{3}{2}\sin 4\theta - \frac{1}{6}\sin 6\theta\right]_0^{\pi/4} \end{aligned}$$

Now substitute the limits:

$$\sin\frac{\pi}{2} = 1, \qquad \sin\pi = 0, \qquad \sin\frac{3\pi}{2} = -1$$

so

$$\begin{aligned} I &= \frac{1}{32}\left(10\cdot\frac{\pi}{4} - \frac{15}{2}(1) + \frac{3}{2}(0) - \frac{1}{6}(-1)\right) \\ &= \frac{1}{32}\left(\frac{5\pi}{2} - \frac{15}{2} + \frac{1}{6}\right) \\ &= \frac{1}{32}\left(\frac{15\pi - 44}{6}\right) \\ &= \frac{15\pi - 44}{192} \end{aligned}$$

Hence the exact value is

$$\int_0^1 \frac{x^6}{(1+x^2)^4}\,\mathrm{d}x = \frac{15\pi - 44}{192}$$

9. **(a)** Use de Moivre's theorem to show that

$$2\cos 5\theta = 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta$$ **[4]**

(b) Hence solve the equation

$$x^5 - 5x^3 + 5x + 1 = 0$$

giving all roots in the form $2\cos q\pi$, where q is a rational number. **[4]**

Solution

(a) By de Moivre's theorem,

$$(\cos\theta + i\sin\theta)^5 = \cos 5\theta + i\sin 5\theta$$

Expand the left-hand side using the binomial theorem:

$$\begin{aligned}(\cos\theta + i\sin\theta)^5 &= \cos^5\theta + 5i\cos^4\theta\sin\theta + 10i^2\cos^3\theta\sin^2\theta \\ &\quad + 10i^3\cos^2\theta\sin^3\theta + 5i^4\cos\theta\sin^4\theta + i^5\sin^5\theta \\ &= \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta \\ &\quad - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta\end{aligned}$$

Comparing real parts gives

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$$

Now use

$$\sin^2\theta = 1 - \cos^2\theta \qquad \text{and} \qquad \sin^4\theta = (1-\cos^2\theta)^2$$

So

$$\begin{aligned}\cos 5\theta &= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta) + 5\cos\theta(1-\cos^2\theta)^2 \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta) \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta\end{aligned}$$

Multiplying by 2,

$$2\cos 5\theta = 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta$$

Hence

$$2\cos 5\theta = 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta$$

(b) Let

$$x = 2\cos\theta$$

Then, using part (a),

$$\begin{aligned}x^5 - 5x^3 + 5x &= (2\cos\theta)^5 - 5(2\cos\theta)^3 + 5(2\cos\theta) \\ &= 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta \\ &= 2\cos 5\theta\end{aligned}$$

So the equation

$$x^5 - 5x^3 + 5x + 1 = 0$$

becomes

$$2\cos 5\theta + 1 = 0$$

that is,

$$\cos 5\theta = -\frac{1}{2}$$

To obtain distinct values of $x = 2\cos\theta$, take $0 \leq \theta \leq \pi$. Then $0 \leq 5\theta \leq 5\pi$, and in this interval

$$\cos 5\theta = -\frac{1}{2}$$

when

$$5\theta = \frac{2\pi}{3}, \ \frac{4\pi}{3}, \ \frac{8\pi}{3}, \ \frac{10\pi}{3}, \ \frac{14\pi}{3}$$

Hence

$$\theta = \frac{2\pi}{15}, \ \frac{4\pi}{15}, \ \frac{8\pi}{15}, \ \frac{2\pi}{3}, \ \frac{14\pi}{15}$$

Therefore the roots are

$$x = 2\cos\frac{2\pi}{15}, \quad 2\cos\frac{4\pi}{15}, \quad 2\cos\frac{8\pi}{15}, \quad 2\cos\frac{2\pi}{3}, \quad 2\cos\frac{14\pi}{15}$$

These are five distinct roots, so they are all the roots of the equation.

10. **(a)** If n is a positive integer and $z = \cos\theta + \mathrm{i}\sin\theta$, show that

$$z^n + \frac{1}{z^n} = 2\cos n\theta, \qquad z^n - \frac{1}{z^n} = 2\mathrm{i}\sin n\theta$$

[2]

(b) By considering $\left(z + \frac{1}{z}\right)^3 \left(z - \frac{1}{z}\right)^2$, find constants A, B and C such that

$$\sin^2\theta\cos^3\theta = A\cos 5\theta + B\cos 3\theta + C\cos\theta$$

[6]

Solution

(a) Write

$$z = \cos\theta + \mathrm{i}\sin\theta = e^{\mathrm{i}\theta}$$

Then by De Moivre's theorem,

$$z^n = e^{\mathrm{i}n\theta} = \cos n\theta + \mathrm{i}\sin n\theta$$

Also,

$$\frac{1}{z} = e^{-\mathrm{i}\theta} \quad \text{so} \quad \frac{1}{z^n} = z^{-n} = e^{-\mathrm{i}n\theta} = \cos n\theta - \mathrm{i}\sin n\theta$$

Adding gives

$$\begin{aligned} z^n + \frac{1}{z^n} &= (\cos n\theta + \mathrm{i}\sin n\theta) + (\cos n\theta - \mathrm{i}\sin n\theta) \\ &= 2\cos n\theta \end{aligned}$$

Subtracting gives

$$\begin{aligned} z^n - \frac{1}{z^n} &= (\cos n\theta + \mathrm{i}\sin n\theta) - (\cos n\theta - \mathrm{i}\sin n\theta) \\ &= 2\mathrm{i}\sin n\theta \end{aligned}$$

Hence

$$z^n + \frac{1}{z^n} = 2\cos n\theta, \qquad z^n - \frac{1}{z^n} = 2\mathrm{i}\sin n\theta$$

(b) From part (a), with $n = 1$,

$$z + \frac{1}{z} = 2\cos\theta, \qquad z - \frac{1}{z} = 2\mathrm{i}\sin\theta$$

So

$$\begin{aligned} \left(z + \frac{1}{z}\right)^3 \left(z - \frac{1}{z}\right)^2 &= (2\cos\theta)^3 (2\mathrm{i}\sin\theta)^2 \\ &= 8\cos^3\theta \cdot 4\mathrm{i}^2\sin^2\theta \\ &= 8\cos^3\theta \cdot (-4)\sin^2\theta \\ &= -32\sin^2\theta\cos^3\theta \end{aligned}$$

Now expand the expression algebraically.
First,

$$\left(z + \frac{1}{z}\right)^3 = z^3 + 3z + \frac{3}{z} + \frac{1}{z^3}$$

and

$$\left(z - \frac{1}{z}\right)^2 = z^2 - 2 + \frac{1}{z^2}$$

Hence

$$\begin{aligned}\left(z+\frac{1}{z}\right)^3\left(z-\frac{1}{z}\right)^2 &= \left(z^3+3z+\frac{3}{z}+\frac{1}{z^3}\right)\left(z^2-2+\frac{1}{z^2}\right)\\ &= \left(z^5-2z^3+z\right)+\left(3z^3-6z+\frac{3}{z}\right)\\ &\quad+\left(3z-\frac{6}{z}+\frac{3}{z^3}\right)+\left(\frac{1}{z}-\frac{2}{z^3}+\frac{1}{z^5}\right)\\ &= z^5+z^3-2z-\frac{2}{z}+\frac{1}{z^3}+\frac{1}{z^5}\end{aligned}$$

Group the terms in pairs:

$$\left(z^5+\frac{1}{z^5}\right)+\left(z^3+\frac{1}{z^3}\right)-2\left(z+\frac{1}{z}\right)$$

Using part (a) again,

$$z^5+\frac{1}{z^5}=2\cos 5\theta, \qquad z^3+\frac{1}{z^3}=2\cos 3\theta, \qquad z+\frac{1}{z}=2\cos\theta$$

So

$$\left(z+\frac{1}{z}\right)^3\left(z-\frac{1}{z}\right)^2=2\cos 5\theta+2\cos 3\theta-4\cos\theta$$

Equating the two expressions,

$$-32\sin^2\theta\cos^3\theta=2\cos 5\theta+2\cos 3\theta-4\cos\theta$$

Therefore

$$\begin{aligned}\sin^2\theta\cos^3\theta &= -\frac{1}{32}\left(2\cos 5\theta+2\cos 3\theta-4\cos\theta\right)\\ &= -\frac{1}{16}\cos 5\theta-\frac{1}{16}\cos 3\theta+\frac{1}{8}\cos\theta\end{aligned}$$

So the constants are

$$A=-\frac{1}{16}, \qquad B=-\frac{1}{16}, \qquad C=\frac{1}{8}$$

11. **(a)** By writing $1 + \mathrm{i}\tan\theta = \sec\theta(\cos\theta + \mathrm{i}\sin\theta)$ and using de Moivre's theorem, show that

$$(1 + \mathrm{i}\tan\theta)^4 = \frac{\cos 4\theta + \mathrm{i}\sin 4\theta}{\cos^4\theta}$$

[3]

(b) Expand $(1 + \mathrm{i}\tan\theta)^4$ and hence show that

$$\cos 4\theta = \frac{1 - 6\tan^2\theta + \tan^4\theta}{(1 + \tan^2\theta)^2}$$

and

$$\sin 4\theta = \frac{4\tan\theta(1 - \tan^2\theta)}{(1 + \tan^2\theta)^2}$$

[4]

(c) Deduce that

$$\tan 4\theta = \frac{4\tan\theta(1 - \tan^2\theta)}{1 - 6\tan^2\theta + \tan^4\theta}$$

[2]

Solution

(a) Starting with

$$1 + \mathrm{i}\tan\theta = 1 + \mathrm{i}\frac{\sin\theta}{\cos\theta} = \frac{\cos\theta + \mathrm{i}\sin\theta}{\cos\theta} = \sec\theta(\cos\theta + \mathrm{i}\sin\theta)$$

Raise both sides to the 4th power:

$$(1 + \mathrm{i}\tan\theta)^4 = \sec^4\theta(\cos\theta + \mathrm{i}\sin\theta)^4$$

By de Moivre's theorem,

$$(\cos\theta + \mathrm{i}\sin\theta)^4 = \cos 4\theta + \mathrm{i}\sin 4\theta$$

so

$$(1 + \mathrm{i}\tan\theta)^4 = \sec^4\theta(\cos 4\theta + \mathrm{i}\sin 4\theta) = \frac{\cos 4\theta + \mathrm{i}\sin 4\theta}{\cos^4\theta}$$

Hence

$$(1 + \mathrm{i}\tan\theta)^4 = \frac{\cos 4\theta + \mathrm{i}\sin 4\theta}{\cos^4\theta}$$

(b) Let $t = \tan\theta$. Then

$$(1 + \mathrm{i}\tan\theta)^4 = (1 + \mathrm{i}t)^4$$

Expanding:

$$\begin{aligned}
(1 + \mathrm{i}t)^4 &= 1 + 4\mathrm{i}t + 6(\mathrm{i}t)^2 + 4(\mathrm{i}t)^3 + (\mathrm{i}t)^4 \\
&= 1 + 4\mathrm{i}t + 6\mathrm{i}^2t^2 + 4\mathrm{i}^3t^3 + \mathrm{i}^4t^4 \\
&= 1 + 4\mathrm{i}t - 6t^2 - 4\mathrm{i}t^3 + t^4 \\
&= (1 - 6t^2 + t^4) + \mathrm{i}(4t - 4t^3) \\
&= (1 - 6t^2 + t^4) + \mathrm{i}\,4t(1 - t^2)
\end{aligned}$$

From part (a),

$$(1 + \mathrm{i}t)^4 = \frac{\cos 4\theta + \mathrm{i}\sin 4\theta}{\cos^4\theta}$$

Comparing real and imaginary parts:

$$\frac{\cos 4\theta}{\cos^4\theta} = 1 - 6t^2 + t^4 \qquad \text{and} \qquad \frac{\sin 4\theta}{\cos^4\theta} = 4t(1 - t^2)$$

Now

$$1 + \tan^2\theta = \sec^2\theta = \frac{1}{\cos^2\theta}$$

so

$$\cos^4\theta = \frac{1}{(1+\tan^2\theta)^2} = \frac{1}{(1+t^2)^2}$$

Therefore

$$\begin{aligned}\cos 4\theta &= (1 - 6t^2 + t^4)\cos^4\theta \\ &= \frac{1 - 6t^2 + t^4}{(1+t^2)^2}\end{aligned}$$

and

$$\begin{aligned}\sin 4\theta &= 4t(1-t^2)\cos^4\theta \\ &= \frac{4t(1-t^2)}{(1+t^2)^2}\end{aligned}$$

Replacing t by $\tan\theta$ gives

$$\cos 4\theta = \frac{1 - 6\tan^2\theta + \tan^4\theta}{(1+\tan^2\theta)^2}$$

and

$$\sin 4\theta = \frac{4\tan\theta(1-\tan^2\theta)}{(1+\tan^2\theta)^2}$$

(c) Using

$$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$$

and the results from part (b),

$$\begin{aligned}\tan 4\theta &= \frac{\dfrac{4\tan\theta(1-\tan^2\theta)}{(1+\tan^2\theta)^2}}{\dfrac{1 - 6\tan^2\theta + \tan^4\theta}{(1+\tan^2\theta)^2}} \\ &= \frac{4\tan\theta(1-\tan^2\theta)}{1 - 6\tan^2\theta + \tan^4\theta}\end{aligned}$$

Hence

$$\tan 4\theta = \frac{4\tan\theta(1-\tan^2\theta)}{1 - 6\tan^2\theta + \tan^4\theta}$$

12. **(a)** Use de Moivre's theorem to show that

$$\sec 6\theta = \frac{\sec^6\theta}{32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta}$$ **[6]**

(b) Hence obtain the roots of the equation

$$x^6 - 18x^4 + 48x^2 - 32 = 0$$

in the form $\sec(q\pi)$, where q is rational. **[4]**

Solution

(a) By de Moivre's theorem,

$$(\cos\theta + \mathrm{i}\sin\theta)^6 = \cos 6\theta + \mathrm{i}\sin 6\theta$$

Let $c = \cos\theta$ and $s = \sin\theta$. Expanding the left-hand side,

$$(c + \mathrm{i}s)^6 = c^6 + 6c^5(\mathrm{i}s) + 15c^4(\mathrm{i}s)^2 + 20c^3(\mathrm{i}s)^3 + 15c^2(\mathrm{i}s)^4 + 6c(\mathrm{i}s)^5 + (\mathrm{i}s)^6$$

Taking real parts gives

$$\cos 6\theta = c^6 - 15c^4s^2 + 15c^2s^4 - s^6$$

Now use $s^2 = 1 - c^2$:

$$\begin{aligned}\cos 6\theta &= c^6 - 15c^4(1-c^2) + 15c^2(1-c^2)^2 - (1-c^2)^3\\ &= c^6 - 15c^4 + 15c^6 + 15c^2(1 - 2c^2 + c^4) - (1 - 3c^2 + 3c^4 - c^6)\\ &= c^6 - 15c^4 + 15c^6 + 15c^2 - 30c^4 + 15c^6 - 1 + 3c^2 - 3c^4 + c^6\\ &= 32c^6 - 48c^4 + 18c^2 - 1\end{aligned}$$

So

$$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$$

Hence

$$\sec 6\theta = \frac{1}{32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1}$$

Multiply top and bottom by $\sec^6\theta$:

$$\sec 6\theta = \frac{\sec^6\theta}{32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta}$$

Hence,

$$\sec 6\theta = \frac{\sec^6\theta}{32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta}$$

(b) Let

$$x = \sec\theta$$

Then the equation

$$x^6 - 18x^4 + 48x^2 - 32 = 0$$

becomes

$$\sec^6\theta - 18\sec^4\theta + 48\sec^2\theta - 32 = 0$$

or

$$32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta = 0$$

From part (a),

$$\sec 6\theta = \frac{\sec^6\theta}{32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta}$$

So if

$$32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta = 0$$

then $\sec 6\theta$ is undefined. Since $\sec^6\theta \neq 0$, this means

$$\cos 6\theta = 0$$

Therefore

$$6\theta = \frac{(2k+1)\pi}{2}$$

so

$$\theta = \frac{(2k+1)\pi}{12}$$

Hence

$$x = \sec\left(\frac{(2k+1)\pi}{12}\right)$$

Taking $k = 0, 1, 2, 3, 4, 5$ gives the six distinct roots:

$$\sec\frac{\pi}{12}, \quad \sec\frac{\pi}{4}, \quad \sec\frac{5\pi}{12}, \quad \sec\frac{7\pi}{12}, \quad \sec\frac{3\pi}{4}, \quad \sec\frac{11\pi}{12}$$

So the roots are

$$x = \sec\left(\frac{(2k+1)\pi}{12}\right), \qquad k = 0, 1, 2, 3, 4, 5$$

13. **(a)** Use de Moivre's theorem to show that

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$ **[5]**

(b) Hence solve for $0 \le \theta \le \pi$

$$16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = \cos 2\theta$$

giving your answers as exact multiples of π. **[5]**

Solution

(a) Let

$$z = \cos\theta + \mathrm{i}\sin\theta$$

By de Moivre's theorem,

$$z^5 = \cos 5\theta + \mathrm{i}\sin 5\theta$$

Now expand z^5 using the binomial theorem:

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)^5 &= \cos^5\theta + 5\cos^4\theta(\mathrm{i}\sin\theta) + 10\cos^3\theta(\mathrm{i}\sin\theta)^2 \\ &\quad + 10\cos^2\theta(\mathrm{i}\sin\theta)^3 + 5\cos\theta(\mathrm{i}\sin\theta)^4 + (\mathrm{i}\sin\theta)^5 \\ &= \cos^5\theta + 5\mathrm{i}\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta \\ &\quad - 10\mathrm{i}\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + \mathrm{i}\sin^5\theta\end{aligned}$$

Equating real parts gives

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$$

Now use

$$\sin^2\theta = 1 - \cos^2\theta, \qquad \sin^4\theta = (1 - \cos^2\theta)^2$$

So

$$\begin{aligned}\cos 5\theta &= \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - \cos^2\theta)^2 \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta) \\ &= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta\end{aligned}$$

Hence

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$

(b) Using part (a), the equation

$$16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = \cos 2\theta$$

becomes

$$\cos 5\theta = \cos 2\theta$$

For $\cos A = \cos B$, the general solution is

$$A = 2n\pi \pm B \qquad (n \in \mathbb{Z})$$

So

$$5\theta = 2n\pi \pm 2\theta$$

This gives two cases.

From

$$5\theta = 2n\pi + 2\theta$$

we get

$$3\theta = 2n\pi$$

so

$$\theta = \frac{2n\pi}{3}$$

For $0 \leq \theta \leq \pi$, this gives

$$\theta = 0,\ \frac{2\pi}{3}$$

From

$$5\theta = 2n\pi - 2\theta$$

we get

$$7\theta = 2n\pi$$

so

$$\theta = \frac{2n\pi}{7}$$

For $0 \leq \theta \leq \pi$, this gives

$$\theta = 0,\ \frac{2\pi}{7},\ \frac{4\pi}{7},\ \frac{6\pi}{7}$$

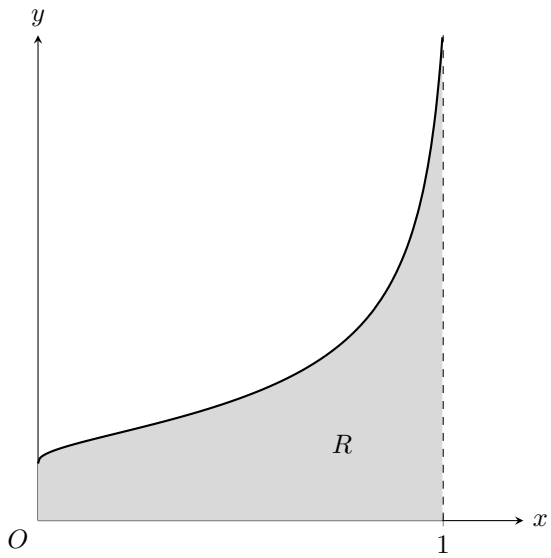
Combining the distinct solutions,

$$\theta = 0,\ \frac{2\pi}{7},\ \frac{4\pi}{7},\ \frac{2\pi}{3},\ \frac{6\pi}{7}$$

These are the required exact values.

14. **(a)** Use de Moivre's theorem to determine constants A, B and C such that

$$\cos^5\theta \equiv A\cos 5\theta + B\cos 3\theta + C\cos\theta$$ **[5]**



The function f is defined by

$$f(x) = -3\sin\left(5\cos^{-1}\left(x^{\frac{1}{6}}\right)\right) - 25\sin\left(3\cos^{-1}\left(x^{\frac{1}{6}}\right)\right) - 150\sin\left(\cos^{-1}\left(x^{\frac{1}{6}}\right)\right), \qquad 0 \le x < 1$$

(b) Show that

$$f'(x) = \frac{40}{\sqrt{1-x^{\frac{1}{3}}}}$$ **[6]**

The diagram shows the curve with equation

$$y = \frac{1}{\sqrt{1-x^{\frac{1}{3}}}}$$

for $0 \le x < 1$ and the asymptote $x = 1$. The region R is the unbounded region between the curve, the x-axis, the line $x = 0$ and the line $x = 1$.
You are given that the area of R is finite.

(c) Determine the exact area of R. **[3]**

Solution

(a) Let $c = \cos\theta$ and $s = \sin\theta$.
Using de Moivre's theorem,

$$(\cos\theta + \mathrm{i}\sin\theta)^5 = \cos 5\theta + \mathrm{i}\sin 5\theta$$

Expanding the left-hand side,

$$\begin{aligned}(c+\mathrm{i}s)^5 &= c^5 + 5c^4(\mathrm{i}s) + 10c^3(\mathrm{i}s)^2 + 10c^2(\mathrm{i}s)^3 + 5c(\mathrm{i}s)^4 + (\mathrm{i}s)^5\\ &= c^5 + 5\mathrm{i}c^4 s - 10c^3s^2 - 10\mathrm{i}c^2s^3 + 5cs^4 + \mathrm{i}s^5\end{aligned}$$

Taking real parts,

$$\cos 5\theta = c^5 - 10c^3s^2 + 5cs^4$$

Since $s^2 = 1 - c^2$,

$$\begin{aligned}\cos 5\theta &= c^5 - 10c^3(1-c^2) + 5c(1-c^2)^2 \\ &= c^5 - 10c^3 + 10c^5 + 5c(1 - 2c^2 + c^4) \\ &= 16c^5 - 20c^3 + 5c\end{aligned}$$

Similarly, from

$$(\cos\theta + \mathrm{i}\sin\theta)^3 = \cos 3\theta + \mathrm{i}\sin 3\theta$$

we get

$$\begin{aligned}(c + \mathrm{i}s)^3 &= c^3 + 3c^2(\mathrm{i}s) + 3c(\mathrm{i}s)^2 + (\mathrm{i}s)^3 \\ &= c^3 + 3\mathrm{i}c^2 s - 3cs^2 - \mathrm{i}s^3\end{aligned}$$

so taking real parts,

$$\begin{aligned}\cos 3\theta &= c^3 - 3cs^2 \\ &= c^3 - 3c(1-c^2) \\ &= 4c^3 - 3c\end{aligned}$$

Now use

$$c^5 \equiv A\cos 5\theta + B\cos 3\theta + C\cos\theta$$

Substituting the expansions,

$$\begin{aligned}c^5 &\equiv A(16c^5 - 20c^3 + 5c) + B(4c^3 - 3c) + Cc \\ &\equiv 16Ac^5 + (-20A + 4B)c^3 + (5A - 3B + C)c\end{aligned}$$

Equating coefficients of powers of c,

$$\begin{aligned}16A &= 1 \\ -20A + 4B &= 0 \\ 5A - 3B + C &= 0\end{aligned}$$

So

$$\begin{aligned}A &= \frac{1}{16} \\ -20\left(\frac{1}{16}\right) + 4B = 0 &\implies 4B = \frac{5}{4} \implies B = \frac{5}{16} \\ 5\left(\frac{1}{16}\right) - 3\left(\frac{5}{16}\right) + C = 0 &\implies -\frac{10}{16} + C = 0 \implies C = \frac{10}{16} = \frac{5}{8}\end{aligned}$$

Hence

$$A = \frac{1}{16}, \qquad B = \frac{5}{16}, \qquad C = \frac{5}{8}$$

(b) Let

$$\theta = \cos^{-1}\left(x^{1/6}\right)$$

so that

$$\cos\theta = x^{1/6}$$

Then

$$f(x) = -3\sin 5\theta - 25\sin 3\theta - 150\sin\theta$$

Differentiate using the chain rule:

$$\begin{aligned}f'(x) &= -3(5\cos 5\theta)\frac{\mathrm{d}\theta}{\mathrm{d}x} - 25(3\cos 3\theta)\frac{\mathrm{d}\theta}{\mathrm{d}x} - 150(\cos\theta)\frac{\mathrm{d}\theta}{\mathrm{d}x} \\ &= -(15\cos 5\theta + 75\cos 3\theta + 150\cos\theta)\frac{\mathrm{d}\theta}{\mathrm{d}x}\end{aligned}$$

Now

$$\theta = \cos^{-1}(x^{1/6})$$

and

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\cos^{-1} u\right) = -\frac{1}{\sqrt{1-u^2}}\frac{\mathrm{d}u}{\mathrm{d}x}$$

Here $u = x^{1/6}$, so

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{6}x^{-5/6}$$

and therefore

$$\begin{aligned}\frac{\mathrm{d}\theta}{\mathrm{d}x} &= -\frac{1}{\sqrt{1-(x^{1/6})^2}} \cdot \frac{1}{6}x^{-5/6} \\ &= -\frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}}\end{aligned}$$

Substitute this into the expression for $f'(x)$:

$$\begin{aligned}f'(x) &= -(15\cos 5\theta + 75\cos 3\theta + 150\cos\theta)\left(-\frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}}\right) \\ &= \frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}}(15\cos 5\theta + 75\cos 3\theta + 150\cos\theta) \\ &= \frac{5}{2}\cdot\frac{x^{-5/6}}{\sqrt{1-x^{1/3}}}\left(\cos 5\theta + 5\cos 3\theta + 10\cos\theta\right)\end{aligned}$$

From part (a),

$$\cos^5\theta = \frac{1}{16}\cos 5\theta + \frac{5}{16}\cos 3\theta + \frac{5}{8}\cos\theta$$

Multiply by 16:

$$16\cos^5\theta = \cos 5\theta + 5\cos 3\theta + 10\cos\theta$$

So

$$f'(x) = \frac{5}{2}\cdot\frac{x^{-5/6}}{\sqrt{1-x^{1/3}}}\cdot 16\cos^5\theta$$

But $\cos\theta = x^{1/6}$, so

$$\cos^5\theta = \left(x^{1/6}\right)^5 = x^{5/6}$$

Hence

$$\begin{aligned}f'(x) &= \frac{5}{2}\cdot\frac{x^{-5/6}}{\sqrt{1-x^{1/3}}}\cdot 16x^{5/6} \\ &= \frac{40}{\sqrt{1-x^{1/3}}}\end{aligned}$$

Therefore

$$f'(x) = \frac{40}{\sqrt{1-x^{1/3}}}$$

(c) The area of R is

$$\int_0^1 \frac{1}{\sqrt{1-x^{1/3}}}\,\mathrm{d}x$$

Since the curve has asymptote $x = 1$, this is an improper integral:

$$\text{Area of } R = \lim_{b\to 1^-}\int_0^b \frac{1}{\sqrt{1-x^{1/3}}}\,\mathrm{d}x$$

From part (b),

$$\frac{1}{\sqrt{1-x^{1/3}}} = \frac{1}{40}f'(x)$$

So

$$\begin{aligned}\text{Area of } R &= \frac{1}{40}\lim_{b\to 1^-}\int_0^b f'(x)\,\mathrm{d}x \\ &= \frac{1}{40}\lim_{b\to 1^-}\big[f(x)\big]_0^b \\ &= \frac{1}{40}\left(\lim_{b\to 1^-} f(b) - f(0)\right)\end{aligned}$$

First find $\lim\limits_{b\to 1^-} f(b)$.

As $b \to 1^-$,

$$b^{1/6} \to 1 \qquad \text{so} \qquad \cos^{-1}(b^{1/6}) \to 0$$

and therefore

$$f(b) \to -3\sin 0 - 25\sin 0 - 150\sin 0 = 0$$

Now find $f(0)$:

$$\cos^{-1}(0^{1/6}) = \cos^{-1}(0) = \frac{\pi}{2}$$

so

$$\begin{aligned}f(0) &= -3\sin\left(5\cdot\frac{\pi}{2}\right) - 25\sin\left(3\cdot\frac{\pi}{2}\right) - 150\sin\left(\frac{\pi}{2}\right) \\ &= -3(1) - 25(-1) - 150(1) \\ &= -3 + 25 - 150 \\ &= -128\end{aligned}$$

Hence

$$\begin{aligned}\text{Area of } R &= \frac{1}{40}\big(0-(-128)\big) \\ &= \frac{128}{40} \\ &= \frac{16}{5}\end{aligned}$$

The exact area of R is $\dfrac{16}{5}$.

15. **(i)** Use de Moivre's theorem to show that

$$\cos 5\theta \equiv \cos\theta\left(1 - 12\sin^2\theta + 16\sin^4\theta\right)$$ **[5]**

(ii) Hence determine the range of values of

$$\frac{\cos 5\theta}{\cos\theta}$$

given that $\cos\theta \neq 0$. **[7]**

Solution

(i) Using de Moivre's theorem,

$$(\cos\theta + \mathrm{i}\sin\theta)^5 = \cos 5\theta + \mathrm{i}\sin 5\theta$$

Expand the left-hand side by the binomial theorem:

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)^5 &= \cos^5\theta + 5\cos^4\theta(\mathrm{i}\sin\theta) + 10\cos^3\theta(\mathrm{i}\sin\theta)^2 \\ &\quad + 10\cos^2\theta(\mathrm{i}\sin\theta)^3 + 5\cos\theta(\mathrm{i}\sin\theta)^4 + (\mathrm{i}\sin\theta)^5\end{aligned}$$

Now use

$$\mathrm{i}^2 = -1, \qquad \mathrm{i}^3 = -\mathrm{i}, \qquad \mathrm{i}^4 = 1, \qquad \mathrm{i}^5 = \mathrm{i}$$

So

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)^5 &= \cos^5\theta + 5\mathrm{i}\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta \\ &\quad - 10\mathrm{i}\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + \mathrm{i}\sin^5\theta\end{aligned}$$

Equating real parts gives

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$$

Factor out $\cos\theta$:

$$\cos 5\theta = \cos\theta\left(\cos^4\theta - 10\cos^2\theta\sin^2\theta + 5\sin^4\theta\right)$$

Now use $\cos^2\theta = 1 - \sin^2\theta$:

$$\begin{aligned}\cos 5\theta &= \cos\theta\left((1-\sin^2\theta)^2 - 10(1-\sin^2\theta)\sin^2\theta + 5\sin^4\theta\right) \\ &= \cos\theta\left(1 - 2\sin^2\theta + \sin^4\theta - 10\sin^2\theta + 10\sin^4\theta + 5\sin^4\theta\right) \\ &= \cos\theta\left(1 - 12\sin^2\theta + 16\sin^4\theta\right)\end{aligned}$$

Hence

$$\cos 5\theta \equiv \cos\theta\left(1 - 12\sin^2\theta + 16\sin^4\theta\right)$$

(ii) From part (i),

$$\frac{\cos 5\theta}{\cos\theta} = 1 - 12\sin^2\theta + 16\sin^4\theta$$

provided $\cos\theta \neq 0$.

Let

$$u = \sin^2\theta$$

Then, since $0 \leq \sin^2\theta \leq 1$ and $\cos\theta \neq 0$, we have

$$0 \leq u < 1$$

So we need the range of

$$f(u) = 1 - 12u + 16u^2 \qquad \text{for } 0 \leq u < 1$$

This is a quadratic opening upwards, so its minimum occurs at the vertex:

$$u = \frac{-(-12)}{2\cdot 16} = \frac{12}{32} = \frac{3}{8}$$

The minimum value is

$$\begin{aligned} f\left(\frac{3}{8}\right) &= 1 - 12\left(\frac{3}{8}\right) + 16\left(\frac{3}{8}\right)^2 \\ &= 1 - \frac{36}{8} + 16 \cdot \frac{9}{64} \\ &= 1 - \frac{9}{2} + \frac{9}{4} \\ &= \frac{4}{4} - \frac{18}{4} + \frac{9}{4} \\ &= -\frac{5}{4} \end{aligned}$$

So the least possible value is

$$-\frac{5}{4}$$

Now consider the largest possible value on $0 \leq u < 1$.
As $u \to 1^-$,

$$f(u) \to 1 - 12 + 16 = 5$$

We must check whether 5 is actually attained. Solve

$$\begin{aligned} 1 - 12u + 16u^2 &= 5 \\ 16u^2 - 12u - 4 &= 0 \\ 4u^2 - 3u - 1 &= 0 \\ (4u+1)(u-1) &= 0 \end{aligned}$$

So

$$u = 1 \quad \text{or} \quad u = -\frac{1}{4}$$

But $u = \sin^2\theta$ cannot be negative, and $u = 1$ would mean

$$\sin^2\theta = 1 \implies \cos\theta = 0$$

which is excluded.
So 5 is not attained.
Therefore the range of

$$\frac{\cos 5\theta}{\cos\theta}$$

is

$$-\frac{5}{4} \leq \frac{\cos 5\theta}{\cos\theta} < 5$$

16. **(a)** Use de Moivre's theorem to show that

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$ **[5]**

(b) Use the identity given in part (a) to find the 2 positive roots of

$$64x^5 - 80x^3 + 20x + 1 = 0$$

giving your answers to 3 significant figures. **[4]**

Solution

(a) Let $c = \cos\theta$ and $s = \sin\theta$. By de Moivre's theorem,

$$(c + \mathrm{i}s)^5 = \cos 5\theta + \mathrm{i}\sin 5\theta$$

Expanding the left-hand side,

$$\begin{aligned}(c + \mathrm{i}s)^5 &= c^5 + 5c^4(\mathrm{i}s) + 10c^3(\mathrm{i}s)^2 + 10c^2(\mathrm{i}s)^3 + 5c(\mathrm{i}s)^4 + (\mathrm{i}s)^5 \\ &= c^5 + 5\mathrm{i}c^4 s - 10c^3 s^2 - 10\mathrm{i}c^2 s^3 + 5cs^4 + \mathrm{i}s^5\end{aligned}$$

Equating real parts gives

$$\cos 5\theta = c^5 - 10c^3 s^2 + 5cs^4$$

Now use $s^2 = 1 - c^2$ and $s^4 = (1 - c^2)^2$:

$$\begin{aligned}\cos 5\theta &= c^5 - 10c^3(1 - c^2) + 5c(1 - c^2)^2 \\ &= c^5 - 10c^3 + 10c^5 + 5c(1 - 2c^2 + c^4) \\ &= c^5 - 10c^3 + 10c^5 + 5c - 10c^3 + 5c^5 \\ &= 16c^5 - 20c^3 + 5c\end{aligned}$$

Since $c = \cos\theta$,

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$

as required.

(b) Let

$$x = \cos\theta \qquad \text{with } 0^\circ \le \theta \le 180^\circ$$

Then

$$\begin{aligned}64x^5 - 80x^3 + 20x + 1 &= 0 \\ 4(16x^5 - 20x^3 + 5x) + 1 &= 0\end{aligned}$$

Using the identity from part (a),

$$16x^5 - 20x^3 + 5x = \cos 5\theta$$

so

$$4\cos 5\theta + 1 = 0$$

Hence

$$\cos 5\theta = -\frac{1}{4}$$

Let

$$\alpha = \arccos\left(-\frac{1}{4}\right) \approx 104.478^\circ$$

Since $0^\circ \le \theta \le 180^\circ$, we have $0^\circ \le 5\theta \le 900^\circ$. Therefore the solutions for 5θ in this interval are

$$\alpha, \quad 360^\circ - \alpha, \quad 360^\circ + \alpha, \quad 720^\circ - \alpha, \quad 720^\circ + \alpha$$

So

$$\theta \approx 20.896^\circ,\ 51.104^\circ,\ 92.896^\circ,\ 123.104^\circ,\ 164.896^\circ$$

Now $x = \cos\theta$, and $x > 0$ when $0^\circ < \theta < 90^\circ$. So only the first two values give positive roots:

$$\begin{aligned}x &= \cos 20.896^\circ \approx 0.934 \\ x &= \cos 51.104^\circ \approx 0.628\end{aligned}$$

So the two positive roots are $x \approx 0.628$ and $x \approx 0.934$.

17. **(a)** Use de Moivre's theorem to show that

$$\sin^2\theta\cos^2\theta = \frac{1}{8}(1-\cos 4\theta)$$

[5]

(b) Hence, or otherwise, find the solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} - 2y\tan\theta = \sin^2\theta$$

for which $y = 0$ when $\theta = \frac{1}{4}\pi$. **[6]**

Solution

(a) By de Moivre's theorem,

$$(\cos\theta + \mathrm{i}\sin\theta)^4 = \cos 4\theta + \mathrm{i}\sin 4\theta$$

Expanding the left-hand side,

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)^4 &= \cos^4\theta + 4\cos^3\theta(\mathrm{i}\sin\theta) + 6\cos^2\theta(\mathrm{i}\sin\theta)^2 \\ &\quad + 4\cos\theta(\mathrm{i}\sin\theta)^3 + (\mathrm{i}\sin\theta)^4 \\ &= \cos^4\theta + 4\mathrm{i}\cos^3\theta\sin\theta - 6\cos^2\theta\sin^2\theta \\ &\quad - 4\mathrm{i}\cos\theta\sin^3\theta + \sin^4\theta\end{aligned}$$

So, comparing real parts,

$$\cos 4\theta = \cos^4\theta - 6\sin^2\theta\cos^2\theta + \sin^4\theta$$

Now rewrite the right-hand side as

$$\begin{aligned}\cos^4\theta - 6\sin^2\theta\cos^2\theta + \sin^4\theta &= (\sin^2\theta + \cos^2\theta)^2 - 8\sin^2\theta\cos^2\theta \\ &= 1 - 8\sin^2\theta\cos^2\theta\end{aligned}$$

Hence

$$\cos 4\theta = 1 - 8\sin^2\theta\cos^2\theta$$

Therefore,

$$\sin^2\theta\cos^2\theta = \frac{1-\cos 4\theta}{8}$$

So

$$\sin^2\theta\cos^2\theta = \frac{1}{8}(1-\cos 4\theta)$$

(b) The differential equation is

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} - 2y\tan\theta = \sin^2\theta$$

This is a linear first order differential equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} + P(\theta)y = Q(\theta)$$

with

$$P(\theta) = -2\tan\theta$$

The integrating factor is

$$\mu(\theta) = e^{\int -2\tan\theta\,\mathrm{d}\theta}$$

Using

$$\int \tan\theta\,\mathrm{d}\theta = -\ln|\cos\theta|$$

we get

$$\mu(\theta) = e^{2\ln|\cos\theta|} = \cos^2\theta$$

Multiply the differential equation through by $\cos^2\theta$:

$$\cos^2\theta\frac{\mathrm{d}y}{\mathrm{d}\theta} - 2y\sin\theta\cos\theta = \sin^2\theta\cos^2\theta$$

The left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}\theta}(y\cos^2\theta)$$

So

$$\frac{\mathrm{d}}{\mathrm{d}\theta}(y\cos^2\theta) = \sin^2\theta\cos^2\theta$$

Using part (a),

$$\frac{\mathrm{d}}{\mathrm{d}\theta}(y\cos^2\theta) = \frac{1-\cos 4\theta}{8}$$

Integrating,

$$\begin{aligned} y\cos^2\theta &= \int\frac{1-\cos 4\theta}{8}\,\mathrm{d}\theta \\ &= \frac{1}{8}\int 1\,\mathrm{d}\theta - \frac{1}{8}\int\cos 4\theta\,\mathrm{d}\theta \\ &= \frac{\theta}{8} - \frac{\sin 4\theta}{32} + C \end{aligned}$$

Now use the condition $y = 0$ when $\theta = \frac{\pi}{4}$:

$$0\cdot\cos^2\frac{\pi}{4} = \frac{1}{8}\cdot\frac{\pi}{4} - \frac{\sin\pi}{32} + C$$

So

$$0 = \frac{\pi}{32} + C$$

and hence

$$C = -\frac{\pi}{32}$$

Therefore,

$$y\cos^2\theta = \frac{\theta}{8} - \frac{\sin 4\theta}{32} - \frac{\pi}{32}$$

So

$$y = \frac{\frac{4\theta}{32} - \frac{\sin 4\theta}{32} - \frac{\pi}{32}}{\cos^2\theta} = \frac{4\theta - \sin 4\theta - \pi}{32\cos^2\theta}$$

Thus the required solution is

$$y = \frac{4\theta - \sin 4\theta - \pi}{32\cos^2\theta}$$

18. **(a)** Use de Moivre's theorem to show that

$$\cos^6\theta = a\cos 6\theta + b\cos 4\theta + c\cos 2\theta + d$$

where a, b, c and d are constants to be found. **[5]**

(b) Hence show that

$$\int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta = \frac{11}{48} + \frac{5\pi}{64}$$ **[5]**

Solution

(a) Let

$$z = e^{\mathrm{i}\theta}$$

Then

$$\cos\theta = \frac{z + z^{-1}}{2}$$

and, by de Moivre's theorem,

$$z^n + z^{-n} = e^{\mathrm{i}n\theta} + e^{-\mathrm{i}n\theta} = 2\cos n\theta$$

Now

$$64\cos^6\theta = (z + z^{-1})^6$$

Expanding by the binomial theorem,

$$(z + z^{-1})^6 = z^6 + 6z^4 + 15z^2 + 20 + 15z^{-2} + 6z^{-4} + z^{-6}$$

Group the terms in pairs:

$$\begin{aligned} 64\cos^6\theta &= (z^6 + z^{-6}) + 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) + 20 \\ &= 2\cos 6\theta + 12\cos 4\theta + 30\cos 2\theta + 20 \end{aligned}$$

Dividing by 64,

$$\begin{aligned} \cos^6\theta &= \frac{1}{32}\cos 6\theta + \frac{12}{64}\cos 4\theta + \frac{30}{64}\cos 2\theta + \frac{20}{64} \\ &= \frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta + \frac{15}{32}\cos 2\theta + \frac{5}{16} \end{aligned}$$

Hence

$$a = \frac{1}{32}, \qquad b = \frac{3}{16}, \qquad c = \frac{15}{32}, \qquad d = \frac{5}{16}$$

(b) Using the identity from part (a),

$$\cos^6\theta = \frac{1}{32}\big(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10\big)$$

So

$$\begin{aligned} \int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta &= \frac{1}{32}\int_0^{\pi/4} \big(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10\big)\,\mathrm{d}\theta \\ &= \frac{1}{32}\left[\frac{1}{6}\sin 6\theta + \frac{6}{4}\sin 4\theta + \frac{15}{2}\sin 2\theta + 10\theta\right]_0^{\pi/4} \\ &= \frac{1}{32}\left[\frac{1}{6}\sin 6\theta + \frac{3}{2}\sin 4\theta + \frac{15}{2}\sin 2\theta + 10\theta\right]_0^{\pi/4} \end{aligned}$$

At $\theta = \frac{\pi}{4}$,

$$\sin\frac{3\pi}{2} = -1, \qquad \sin\pi = 0, \qquad \sin\frac{\pi}{2} = 1$$

Therefore

$$\begin{aligned}\int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta &= \frac{1}{32}\left(-\frac{1}{6}+\frac{3}{2}\cdot 0+\frac{15}{2}\cdot 1+10\cdot\frac{\pi}{4}\right)\\ &= \frac{1}{32}\left(-\frac{1}{6}+\frac{15}{2}+\frac{5\pi}{2}\right)\\ &= \frac{1}{32}\left(\frac{44}{6}+\frac{5\pi}{2}\right)\\ &= \frac{11}{48}+\frac{5\pi}{64}\end{aligned}$$

Hence

$$\int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta = \frac{11}{48}+\frac{5\pi}{64}$$